

TASK # 01 — Temperature Status (if-else)

A weather monitoring system takes the temperature in Celsius from the user. Write a C program using if-else to display “Cold” if the temperature is below 15, “Normal” if the temperature is between 15 and 30, and “Hot” if the temperature is above 30.

```
postlab5 Q1.cpp
1  #include<stdio.h>
2  int main()
3  {
4      float celsius;
5      printf("Enter The Temperature = ");
6      scanf("%f", &celsius);
7      if (celsius < 15)
8      {
9          printf("Cold");
10     }
11     else if (celsius >= 15 && celsius <= 30)
12     {
13         printf("Normal");
14     }
15     else if (celsius > 30)
16     {
17         printf("Hot");
18     }
19 }
```

```
C:\Users\Hassan Muhammad x + v
Enter The Temperature = 14
Cold
-----
Process exited after 4.11 seconds with return value 0
Press any key to continue . . . |
```

```
C:\Users\Hassan Muhammad x + v
Enter The Temperature = 40
Hot
-----
Process exited after 2.44 seconds with return value 0
Press any key to continue . . . |
```

TASK # 02 — Mobile Data Package (if-else)

A mobile service provider checks a customer's remaining balance before activating a package. Write a C program that takes the balance as an integer and uses if-else to display "Low Balance" if the balance is below 500, "Sufficient Balance" if it is between 500 and 2000, and "Premium Balance" if it is above 2000.

```
postlab5 Q1.cpp  postlab5 Q2.cpp
1  #include<stdio.h>
2  int main()
3  {
4      int Balance;
5      printf("Enter Your Balance Please = ");
6      scanf("%d", &Balance);
7      if (Balance < 500)
8      {
9          printf("Low Balance");
10     }
11     else if (Balance >= 500 && Balance <= 2000)
12     {
13         printf("Sufficient Balance");
14     }
15     else
16     {
17         printf("Premium Balance");
18     }
19 }
```

```
C:\Users\Hassan Muhammad  X  +  v
Enter Your Balance Please = 600
Sufficient Balance
-----
Process exited after 3.482 seconds with return value 0
Press any key to continue . . .
```

```
C:\Users\Hassan Muhammad  X  +  v
Enter Your Balance Please = 4000
Premium Balance
-----
Process exited after 6.567 seconds with return value 0
Press any key to continue . . .
```

TASK # 03 — Hospital Appointment (nested if-else)

A hospital appointment system asks the patient to enter three integer values: appointment, doctor Available, and registration Completed, where 1 means yes and 0 means no. Write a C program using nested if-else statements to first check whether the patient has an appointment, then check whether the doctor is available, and finally check whether the registration is completed before displaying whether the patient can meet the doctor.

```
postlab5 Q1.cpp  postlab5 Q2.cpp  postlab5 Q3.cpp
1  #include<stdio.h>
2  int main()
3  {
4      int appointment , DrAvailable , registration;
5      printf("Do You Have An Appointment ? 1 For Yes and 2 For No ");
6      scanf("%d", &appointment);
7      printf("Is The Doctor Available ? 1 For Yes and 2 For No ");
8      scanf("%d", &DrAvailable);
9      printf("Is Your Registration Completed ? 1 For Yes and 2 For No ");
10     scanf("%d", &registration);
11     if (appointment == 1)
12     {
13         if (DrAvailable == 1)
14         {
15             if (registration == 1)
16             {
17                 printf("Yes You can meet the Doctor");
18             }
19             else
20             {
21                 printf("Sorry You cannot meet the Doctor");
22             }
23         }
24         else
25         {
26             printf("Sorry You cannot meet the Doctor");
27         }
28     }
29     else
30     {
31         printf("Sorry You cannot meet the Doctor");
32     }
33 }
```

```
C:\Users\Hassan Muhammad x + v
Do You Have An Appointment ? 1 For Yes and 2 For No 1
Is The Doctor Available ? 1 For Yes and 2 For No 1
Is Your Registration Completed ? 1 For Yes and 2 For No 1
Yes You can meet the Doctor
-----
Process exited after 3.062 seconds with return value 0
Press any key to continue . . .
```

```
C:\Users\Hassan Muhammad x + v
Do You Have An Appointment ? 1 For Yes and 2 For No 1
Is The Doctor Available ? 1 For Yes and 2 For No 1
Is Your Registration Completed ? 1 For Yes and 2 For No 2
Sorry You cannot meet the Doctor
-----
Process exited after 3.98 seconds with return value 0
Press any key to continue . . . |
```

TASK # 04 — Online Food Order (nested if-else)

An online food delivery system asks the customer to enter three integer values: restaurantOpen, itemAvailable, and balanceSufficient, where 1 means yes and 0 means no. Write a C program using nested if-else statements to first check whether the restaurant is open, then whether the selected item is available, and finally whether the customer has sufficient balance before displaying the appropriate order status.

```
postlab5 Q1.cpp postlab5 Q2.cpp postlab5 Q3.cpp postlab Q4.cpp
1  #include<stdio.h>
2  int main()
3  {
4      int restaurantOpen , itemAvailable , balanceSufficient;
5      printf("Is Restaurant Open ? 1 For Yes and 2 For No ");
6      scanf("%d", &restaurantOpen);
7      printf("Is The Selected Item Available ? 1 For Yes and 2 For No ");
8      scanf("%d", &itemAvailable);
9      printf("Do You Have Sufficient Balance ? 1 For Yes and 2 For No ");
10     scanf("%d", &balanceSufficient);
11     if (restaurantOpen == 1)
12     {
13         if (itemAvailable == 1)
14         {
15             if (balanceSufficient == 1)
16             {
17                 printf("\nThank You ! We got Your Order and It is Being Prepared. ");
18             }else
19             {
20                 printf("\nSorry We Couldnot Get Enough Information To Place Your Order. ");
21             }
22         }else
23         {
24             printf("\nSorry We Couldnot Get Enough Information To Place Your Order. ");
25         }
26     }else
27     {
28         printf("\nSorry We Couldnot Get Enough Information To Place Your Order. ");
29     }
30 }
31
```

```
C:\Users\Hassan Muhammad x + v
Is Restaurant Open ? 1 For Yes and 2 For No 1
Is The Selected Item Available ? 1 For Yes and 2 For No 1
Do You Have Sufficient Balance ? 1 For Yes and 2 For No 1

Thank You ! We got Your Order and It is Being Prepared.
-----
Process exited after 2.605 seconds with return value 0
Press any key to continue . . .
```

```
C:\Users\Hassan Muhammad x + v
Is Restaurant Open ? 1 For Yes and 2 For No 1
Is The Selected Item Available ? 1 For Yes and 2 For No 2
Do You Have Sufficient Balance ? 1 For Yes and 2 For No 1

Sorry We Couldnot Get Enough Information To Place Your Order.
-----
Process exited after 5.763 seconds with return value 0
Press any key to continue . . .
```

TASK # 05 — ATM and Transaction Selection (Nested switch):

An ATM provides four operations: 1 for Balance Inquiry, 2 for Cash Withdrawal, 3 for Cash Deposit, and 4 for PIN Change. After the user selects an operation, the ATM displays a second menu related to that operation; for example, after selecting Cash Withdrawal, the user can choose 1 for Savings Account or 2 for Current Account. Write a C program using a nested switch statement to take both choices as integer inputs and display the selected operation and account type.

```
postlab5 Q1.cpp postlab5 Q2.cpp postlab5 Q3.cpp postlab Q4.cpp postlab Q5.cpp
1 #include<stdio.h>
2 int main()
3 {
4     int operations , choice ;
5     printf("You Can Choose From Operations Enter The Number For That Operation . \n 1 For Balance Inquiry. \n 2 For Cash Withdrawal. \n 3 For Cash Deposit. \n 4 F
6     scanf("%d", &operations);
7     switch (operations)
8     {
9         case 1:
10            printf("You Have Selected Balance Inquiry. \n Now Enter 1 For Savings Account. \n Enter 2 For Current Account. \n Your Choice = ");
11            scanf("%d", &choice);
12            switch (choice)
13            {
14                case 1:
15                    printf("You Have Selected Balance Inquiry And Savings Account.");
16                    break;
17                case 2:
18                    printf("You Have Selected Balance Inquiry And Current Account.");
19            }
20            break;
21        case 2:
22            printf("You Have Selected Cash Withdrawal = \n Enter 1 For Savings Account. \n Enter 2 For Current Account. \n Your Choice = ");
23            scanf("%d", &choice);
24            switch (choice)
25            {
26                case 1:
27                    printf("You Have Selected Cash Withdrawal And Savings Account.");
28                    break;
29                case 2:
30                    printf("You Have Selected Cash Withdrawal And Current Account.");
31            }
32            break;
33        case 3:
```

```
postlab5 Q1.cpp postlab5 Q2.cpp postlab5 Q3.cpp postlab Q4.cpp postlab Q5.cpp
1
2
3
4
5     ons Enter The Number For That Operation . \n 1 For Balance Inquiry. \n 2 For Cash Withdrawal. \n 3 For Cash Deposit. \n 4 For PIN Change. \n Your operation = ");
6
7
8
9
10    Balance Inquiry. \n Now Enter 1 For Savings Account. \n Enter 2 For Current Account. \n Your Choice = ");
11
12
13
14
15    elected Balance Inquiry And Savings Account.");
16
17
18    elected Balance Inquiry And Current Account.");
19
20
21
22    Cash Withdrawal = \n Enter 1 For Savings Account. \n Enter 2 For Current Account. \n Your Choice = ");
23
24
25
26
27    elected Cash Withdrawal And Savings Account.");
28
29
30    elected Cash Withdrawal And Current Account.");
31
```

```
postlab5 Q1.cpp postlab5 Q2.cpp postlab5 Q3.cpp postlab Q4.cpp postlab Q5.cpp
30 } printf("You Have Selected Cash Withdrawal And Current Account.");
31 }
32 break;
33 case 3:
34 printf("You Have Selected Cash Deposit = \n Enter 1 For Savings Account. \n Enter 2 For Current Account. \n Your Choice = ");
35 scanf("%d", &choice);
36 switch (choice)
37 {
38 case 1:
39 printf("You Have Selected Cash Deposit And Savings Account.");
40 break;
41 case 2:
42 printf("You Have Selected Cash Deposit And Current Account.");
43 }
44 break;
45 case 4:
46 printf("You Have Selected PIN Change = \n Enter 1 For Savings Account. \n Enter 2 For Current Account. \n Your Choice = ");
47 scanf("%d", &choice);
48 switch (choice)
49 {
50 case 1:
51 printf("You Have Selected PIN Change And Savings Account.");
52 break;
53 case 2:
54 printf("You Have Selected PIN Change And Current Account.");
55 }
56 break;
57 default:
58 printf("You Have Entered Wrong Operation. ");
59 }
60 }
61 }
62 }
```

```
C:\Users\Hassan Muhammad x + v
You Can Choose From Operations Enter The Number For That Operation .
1 For Balance Inquiry.
2 For Cash Withdrawal.
3 For Cash Deposit.
4 For PIN Change.
Your operation = 1
You Have Selected Balance Inquiry =
Enter 1 For Savings Account.
Enter 2 For Current Account.
Your Choice = 2
You Have Selected Balance Inquiry And Current Account.
-----
Process exited after 2.736 seconds with return value 0
Press any key to continue . . .
```

```
C:\Users\Hassan Muhammad x + v
You Can Choose From Operations Enter The Number For That Operation .
1 For Balance Inquiry.
2 For Cash Withdrawal.
3 For Cash Deposit.
4 For PIN Change.
Your operation = 2
You Have Selected Cash Withdrawal =
Enter 1 For Savings Account.
Enter 2 For Current Account.
Your Choice = 1
You Have Selected Cash Withdrawal And Savings Account.
-----
Process exited after 2.415 seconds with return value 0
Press any key to continue . . . |
```

```
C:\Users\Hassan Muhammad x + v
You Can Choose From Operations Enter The Number For That Operation .
1 For Balance Inquiry.
2 For Cash Withdrawal.
3 For Cash Deposit.
4 For PIN Change.
Your operation = 5
You Have Entered Wrong Operation.
-----
Process exited after 2.386 seconds with return value 0
Press any key to continue . . .
```